

**Educational Learning Analytics using Data Warehouse, Text Mining,
Decision Tree, K-Means Clustering, KNIME, WEKA and Microsoft
Power BI**

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfilment for the Course of

CSA16 – Data Warehousing and Data Mining

to the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

Submitted by

N Tejaswitha (193271057)

N Vijetha (192311258)

M.Mahathi (192324098)

B.Lakshmi Lohitha (192324182)

Under the Supervision of

Dr. D. Beulah David



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

August 2025



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai-602105



DECLARATION

We, N. Tejaswitha(192371057), N. Vijetha(192311258), M. Mahathi(192324098), B. Lakshmi Lohitha(192324182) of the Computer Science and Engineering, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled **‘EDUCATIONAL LEARNING ANALYTICS USING DATA WAREHOUSE, TEXT MINING DECISION, K-MEANS CLUSTERING, KNIME, WEKA AND MICROSOFT POWER BI’** is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place: Chennai

Date:

Signature of the Students with Names
N Tejaswitha (193271057)
N Vijetha (192311258)
M. Mahathi (192324098)
B. Lakshmi Lohitha (192324182)



SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105



BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**EDUCATIONAL LEARNING ANALYTICS USING DATA WAREHOUSE, TEXT MINING DECISION, K-MEANS CLUSTERING,KNIME,WEKA AND MICROSOFT POWER BI**” has been carried out by N.Tejaswitha(192371057),N.Vijetha(192311258),M.Mahathi(192324098),B.LakshmiLohitha (192324182) under the supervision of **Dr. D. Beulah David** and is submitted in partial fulfilment of the requirements for the current semester of the B. Tech **Computer Science and Engineering** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

SIGNATURE

Dr. John Justin
Program Director

Department of CSE
Saveetha School of Engineering
SIMATS

SIGNATURE

Dr. D. Beulah David
Professor

Department of CSE
Saveetha School of Engineering
SIMATS

Submitted for the Project work Viva-Voce held on _____
.

INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

We would like to express our heartfelt gratitude to all those who supported and guided us throughout the successful completion of our Capstone Project. We are deeply thankful to our respected Founder and Chancellor, Dr. N.M. Veeraiyan, Saveetha Institute of Medical and Technical Sciences, for his constant encouragement and blessings. We also express our sincere thanks to our Pro-Chancellor, Dr. Deepak Nallaswamy Veeraiyan, and our Vice-Chancellor, Dr. S. Suresh Kumar, for their visionary leadership and moral support during the course of this project.

We are truly grateful to our Director, Dr. Ramya Deepak, SIMATS Engineering, for providing us with the necessary resources and a motivating academic environment. Our special thanks to our Principal, Dr. B. Ramesh for granting us access to the institute's facilities and encouraging us throughout the process. We sincerely thank our Head of the Department, **Dr. Rashmitha Khilar** for her continuous support, valuable guidance, and constant motivation.

We are especially indebted to our guide from the Department of CSE, **Dr. D. Beulah David** for their creative suggestions, consistent feedback, and unwavering support during each stage of the project. We also express our gratitude to the Project Coordinators, Review Panel Members (Internal and External), and the entire faculty team for their constructive feedback and valuable inputs that helped improve the quality of our work. Finally, we thank all faculty members, lab technicians, our parents, and friends for their continuous encouragement and support.

Signature With Student Name

N Tejaswitha (193271057)

N Vijetha (192311258)

M.Mahathi (192324098)

B.Lakshmi Lohitha (192324182)

ABSTRACT

The “Educational Learning Analytics using Data Warehouse, Text Mining, Decision Tree, K-Means Clustering, KNIME, WEKA and Microsoft Power BI” project aims to develop an analytical system for collecting, organizing, processing, and analyzing educational data. In modern educational environments, institutions generate a large amount of data related to students, academic performance, attendance, assessments, learning activities, and feedback. Manual analysis of this information can be difficult and time-consuming. The proposed system addresses this problem by using data warehousing and data mining techniques to transform educational data into meaningful knowledge.

The collected data is integrated into a Data Warehouse, where information from different educational sources can be stored in an organized and structured form. Text Mining is used to analyze textual information such as student feedback, comments, and learning-related responses. Decision Tree techniques are applied to classify student performance and identify factors that influence academic outcomes. K-Means Clustering is used to group students with similar learning behavior and academic characteristics.

KNIME and WEKA are used to support data preprocessing, machine learning, classification, clustering, and analytical experimentation. Microsoft Power BI is used to create interactive dashboards and visual reports that help users understand patterns and trends in educational data. The proposed system allows educators and administrators to view student performance, attendance patterns, learning behavior, and analytical results through clear visualizations.

The system can improve educational decision-making, student monitoring, performance analysis, and early identification of learning difficulties. It can also provide a foundation for future improvements such as real-time learning analytics, predictive student performance analysis, personalized learning recommendations, and automated academic alerts. Overall, the project demonstrates the practical application of Data Warehousing, Text Mining, Machine Learning, KNIME, WEKA, and Microsoft Power BI for improving educational analysis and supporting data-driven decision-making.

TABLE OF CONTENTS

Section	Page
Acknowledgement	6
Abstract	5
Chapter 1: Introduction	8
1.1 Background Information	8
1.2 Project Objectives	8
1.3 Significance	9
1.4 Scope	9
1.5 Methodology Overview	9
Chapter 2: Problem Identification and Analysis	10
2.1 Description of the Problem	10
2.2 Evidence of the Problem	10
2.3 Stakeholders	11
2.4 Supporting Data and Analysis	11
Chapter 3: Solution Design and Implementation	12
3.1 Development and Design Process	12
3.2 Tools and Technologies Used	12
3.3 Solution Overview	13
3.4 Analytical and Data Management Principles	13
3.5 Solution Justification	14
Chapter 4: Results and Recommendations	14
4.1 Evaluation of Results	14
4.2 Challenges Encountered	15
4.3 Possible Improvements	15
4.4 Recommendations	15
Chapter 5: Reflection of Learning and Personal Development	16
5.1 Key Learning Outcomes	16
5.1.1 Academic Knowledge	16
5.1.2 Technical Skills	16
5.1.3 Problem-Solving and Critical Thinking	16
5.2 Challenges Encountered and Overcome	17
5.2.1 Personal and Professional Growth	17
5.2.2 Collaboration and Communication	17
5.3 Application of Technical Principles	17
5.4 Insights into the Industry	17
5.5 Conclusion of Personal Development	18
Chapter 6: Conclusion	18
6.1 Summary of Key Findings	18
6.2 Impacts and Significance	18
6.3 Future Prospects	18
References	

CHAPTER 1 – INTRODUCTION

1.1 Background Information

Modern educational institutions generate large volumes of data through student records, examinations, attendance systems, online learning platforms, assignments, assessments, and feedback. As the amount of educational information increases, it becomes difficult for teachers and administrators to manually analyze student performance and learning behavior. Traditional methods of educational analysis may require significant time and may not easily identify hidden patterns within large datasets.

Educational Learning Analytics provides a systematic approach for collecting, processing, analyzing, and visualizing educational data. A Data Warehouse helps integrate information from multiple sources into a centralized system. Text Mining can analyze student feedback and other textual data to identify important learning-related information. Machine learning techniques such as Decision Tree and K-Means Clustering can identify performance patterns and group students based on similar characteristics.

The proposed project, “Educational Learning Analytics using Data Warehouse, Text Mining, Decision Tree, K-Means Clustering, KNIME, WEKA and Microsoft Power BI,” focuses on developing an analytical approach for understanding educational data. The system collects relevant student information, processes the data, applies analytical techniques, and presents meaningful results through interactive dashboards and visual reports.

1.2 Project Objectives

The main objectives of this project are:

- To understand the concepts of Educational Learning Analytics.
- To collect and organize educational data from different sources.
- To develop a Data Warehouse for storing and managing educational information.
- To perform data preprocessing and transformation.
- To apply Text Mining techniques to educational feedback and textual data.
- To use Decision Tree algorithms for student performance classification.
- To apply K-Means Clustering for grouping students with similar learning characteristics.
- To use KNIME and WEKA for data analysis and machine learning.
- To create interactive dashboards using Microsoft Power BI.
- To improve educational decision-making through data-driven analysis.

1.3 Significance

Educational Learning Analytics is important because educational institutions need accurate information

about student performance and learning behavior. Manual analysis of academic records can be time-consuming and may not identify important patterns. The proposed system helps address this problem by organizing educational data and applying analytical techniques.

The project can help educators understand student performance, identify learning difficulties, analyze attendance patterns, and study feedback. Decision Tree models can support student classification, while K-Means Clustering can identify groups of students with similar characteristics. Microsoft Power BI dashboards can provide a simple and visual representation of the analytical results.

1.4 Scope

This project focuses on the collection, storage, processing, analysis, and visualization of educational data. The system processes information related to students and applies analytical techniques to identify meaningful patterns.

The project includes:

- Student data collection.
- Academic performance analysis.
- Attendance analysis.
- Data preprocessing and cleaning.
- Data Warehouse development.
- Text Mining of student feedback.
- Decision Tree classification.
- K-Means Clustering.
- Analysis using KNIME and WEKA.
- Dashboard development using Microsoft Power BI.
- Visualization of educational patterns and trends.
- Evaluation of analytical results.

1.5 Methodology Overview

The methodology adopted in this project consists of several stages. First, educational data is collected from available sources such as student records, attendance information, examination results, assessments, and feedback. The collected information is cleaned, transformed, and prepared for analysis.

Next, the processed data is stored in a Data Warehouse for organized management and integration. Text Mining techniques are applied to analyze textual information and identify meaningful patterns from feedback and comments. Decision Tree algorithms are used to classify student performance, while K-Means Clustering is used to group students according to similar academic and learning characteristics.

KNIME and WEKA are used to perform data preprocessing, classification, clustering, and model evaluation. The analytical results are then imported into Microsoft Power BI, where dashboards and reports are created.

Finally, the developed system is evaluated based on data processing, classification, clustering, visualization, and usability.

CHAPTER 2 – PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the Problem

Educational institutions manage large amounts of student-related information. This data may include marks, attendance, assessments, learning activities, feedback, and demographic information. As the amount of data increases, manually analyzing every student's academic performance becomes difficult and time-consuming. Teachers and administrators may not always have a complete understanding of student learning patterns. Some students may require additional academic support, while others may show consistent improvement or high performance. Without proper data analysis, identifying these patterns can be challenging.

The proposed project addresses this problem by developing an Educational Learning Analytics system using Data Warehouse, Text Mining, Decision Tree, K-Means Clustering, KNIME, WEKA, and Microsoft Power BI. The system organizes educational data, applies analytical methods, and presents useful information for educational decision-making.

2.2 Evidence of the Problem

The increasing availability of digital educational systems provides clear evidence of the need for effective educational data analysis. Student information is generated continuously through academic and learning activities.

An effective Educational Learning Analytics system can provide:

- Improved understanding of student performance.
- Faster identification of learning patterns.
- Reduced manual analysis of academic records.
- Better management of educational data.
- Identification of students requiring additional support.
- Classification of academic performance.
- Grouping of students with similar characteristics.
- Improved educational decision-making.

During evaluation, the analytical system can be tested by comparing predicted or classified student performance with available academic data. The results can be used to measure classification accuracy, clustering quality, data processing performance, and visualization effectiveness.

2.3 Stakeholders

The following stakeholders are affected by this problem:

- Students: Benefit from improved identification of learning needs and academic performance patterns.
- Teachers: Can use analytical information to understand student progress and provide suitable support.
- Educational Administrators: Require organized information for academic planning and decision-making.
- Educational Institutions: Benefit from improved student monitoring and data management.
- Data Analysts: Can use the system to analyze educational datasets and identify meaningful patterns.
- Researchers and Students: Can study the practical applications of Data Mining and Learning Analytics.
- Parents: May indirectly benefit from improved academic monitoring and student support.

2.4 Supporting Data and Analysis

Educational data analysis is supported by the principles of Data Warehousing, Data Mining, Machine Learning, and Business Intelligence. A Data Warehouse provides centralized storage for information collected from multiple sources. Data Mining techniques help identify hidden patterns and relationships within educational datasets.

The proposed project is supported by the following principles:

- Data Warehousing supports organized educational data storage.
- Data preprocessing improves data quality before analysis.
- Text Mining helps analyze student feedback and textual information.
- Decision Tree supports classification and performance prediction.
- K-Means Clustering groups students with similar characteristics.
- KNIME supports workflow-based data analysis.
- WEKA provides machine learning and data mining techniques.
- Microsoft Power BI supports interactive visualization and reporting.

The analysis demonstrates that combining these technologies can provide a clearer understanding of student learning behavior and academic performance.

CHAPTER 3 – SOLUTION DESIGN AND IMPLEMENTATION

3.1 Development and Design Process

The proposed solution was developed to analyze educational data using Data Warehousing, Text Mining, Machine Learning, and Business Intelligence techniques. First, educational data is collected from available academic sources. The data is then cleaned and transformed to remove errors, missing values, and inconsistencies.

The processed information is stored in a Data Warehouse. Textual data such as student comments and feedback is processed using Text Mining techniques. Decision Tree algorithms are applied to classify students based on academic performance. K-Means Clustering is used to group students according to similar learning and performance characteristics.

KNIME and WEKA are used to develop and test analytical workflows and machine learning models. The final results are visualized using Microsoft Power BI.

Educational Data → Data Preprocessing → Data Warehouse → Text Mining → Decision Tree → K-Means Clustering → Analysis → Power BI Dashboard

3.2 Tools and Technologies Used

The following tools and technologies can be used in this project:

- Data Warehouse – Centralized storage and management of educational data.
- Text Mining – Analysis of student feedback and textual information.
- Decision Tree – Classification of student academic performance.
- K-Means Clustering – Grouping of students based on similar characteristics.
- KNIME – Data analytics and workflow development.
- WEKA – Machine Learning and Data Mining analysis.
- Microsoft Power BI – Interactive dashboards and data visualization.
- Spreadsheet or CSV datasets – Storage and preparation of educational information.
- Database systems – Management of structured educational records.

3.3 Solution Overview

The proposed system provides an analytical approach for understanding educational data and student learning patterns.

The basic workflow is:

Educational Data → Data Warehouse → Data Preprocessing → Text Mining → Machine Learning Analysis

→ Visualization → Decision Support

Using this approach:

- Educational data is collected from available sources.
- The data is cleaned and transformed.
- Information is stored in a Data Warehouse.
- Textual feedback is analyzed using Text Mining.
- Student performance is classified using Decision Tree.
- Students are grouped using K-Means Clustering.
- KNIME and WEKA are used for analytical processing.
- Microsoft Power BI displays the results through dashboards.
- Educators can understand performance patterns and learning trends.

The system helps educational institutions obtain meaningful insights from large amounts of educational data while reducing manual analysis.

3.4 Analytical and Data Management Principles

The project follows relevant data analysis and educational information management principles, including:

- Proper collection and organization of educational data.
- Data cleaning and preprocessing.
- Responsible handling of student information.
- Machine Learning model evaluation.
- Data Mining and clustering analysis.
- Accurate visualization and reporting.
- Privacy and secure management of educational records.

These principles help ensure that the system is developed and used in an organized, responsible, and analytical manner.

3.5 Solution Justification

The proposed solution is suitable because manually analyzing educational information can be difficult when large numbers of students and academic records are involved. Automating data analysis provides faster and more organized educational insights.

The solution provides:

- Centralized educational data storage.
- Automated data preprocessing.
- Student performance classification.
- Student grouping and segmentation.
- Analysis of textual feedback.

- Interactive visual dashboards.
- Reduced manual analytical effort.
- Improved educational decision-making.

CHAPTER 4 – RESULTS AND RECOMMENDATIONS

4.1 Evaluation of Results

The developed Educational Learning Analytics system was evaluated using educational datasets. The system was tested based on data preprocessing, classification, clustering, text analysis, and dashboard visualization.

Obtained Results:

- Educational Dataset: Successfully processed.
- Data Warehouse: Successfully organized.
- Text Analysis: Meaningful keywords and patterns identified.
- Student Classification: Successfully performed using Decision Tree.
- Student Clustering: Successfully performed using K-Means.
- Analytical Workflows: Successfully developed using KNIME and WEKA.
- Dashboard Generation: Successfully completed using Microsoft Power BI.
- Educational Visualization: Performance and learning patterns displayed.

The results indicate that the proposed system can effectively process educational information and identify useful academic patterns. The dashboards provide a clear representation of student performance and analytical outcomes. The system can therefore improve educational monitoring and reduce the effort required for manual data analysis.

4.2 Challenges Encountered

During the development and testing of the system, several challenges were identified:

- Educational data may contain missing values.
- Data from different sources may have different formats.
- Student records may contain incomplete information.
- Textual feedback may contain unstructured data.
- Selecting suitable attributes for classification can be challenging.
- Determining the appropriate number of clusters requires analysis.
- Large educational datasets may require additional processing time.
- Data privacy must be maintained when handling student information.

These challenges can be addressed through data cleaning, preprocessing, proper attribute selection, validation, and responsible data management.

4.3 Possible Improvements

The effectiveness of the system can be improved through the following measures:

- Implement real-time educational data analysis.
- Add early warning systems for low-performing students.
- Introduce personalized learning recommendations.
- Improve Text Mining techniques for feedback analysis.
- Compare multiple Machine Learning algorithms.
- Improve clustering methods for student segmentation.
- Integrate additional educational data sources.
- Add predictive analytics for future academic performance.
- Provide advanced dashboards and reporting options.

4.4 Recommendations

Based on the results, the following recommendations are proposed:

- Maintain an organized educational Data Warehouse.
- Regularly update student academic information.
- Perform data cleaning before analysis.
- Monitor students who show low academic performance.
- Use Machine Learning results as decision-support information.
- Protect student data and maintain privacy.
- Compare current and previous academic performance.
- Use Power BI dashboards for continuous educational monitoring.
- Integrate additional learning analytics tools in future versions.

Overall, the developed system provides a useful approach for analyzing educational data and supporting improved learning and academic decision-making.

CHAPTER 5 – REFLECTION OF LEARNING AND PERSONAL DEVELOPMENT

5.1 Key Learning Outcomes

5.1.1 Academic Knowledge

This project enhanced my understanding of Educational Learning Analytics, Data Warehousing, Text Mining, Data Mining, Machine Learning, and Business Intelligence. I learned how educational information can be collected, processed, and analyzed to identify useful patterns. The project also helped me understand the importance of data-driven decision-making in education.

5.1.2 Technical Skills

Through this project, I developed technical knowledge in data preprocessing, Data Warehouse concepts, Text Mining, Decision Tree classification, K-Means Clustering, KNIME, WEKA, and Microsoft Power BI. I gained practical experience in organizing data, performing analysis, developing machine learning models, and presenting results through visual dashboards.

5.1.3 Problem-Solving and Critical Thinking

The project strengthened my problem-solving abilities by requiring me to manage different types of educational data. I learned how to identify missing values, prepare datasets, select suitable analytical techniques, and interpret the obtained results. I also learned to evaluate the relationship between student data, academic performance, and learning patterns.

5.2 Challenges Encountered and Overcome

During the project, understanding the integration of multiple analytical tools and techniques was initially challenging. Data quality issues, missing information, unstructured textual data, and the selection of appropriate machine learning methods affected the analysis.

These challenges were addressed through continuous learning, data preprocessing, repeated experimentation, validation of analytical results, and comparison of different techniques.

5.2.1 Personal and Professional Growth

This project improved my confidence in applying Data Science and Machine Learning concepts to an educational problem. It enhanced my analytical thinking, research skills, data interpretation, problem-solving abilities, time management, and technical documentation skills.

5.2.2 Collaboration and Communication

Working on this project improved my communication and teamwork skills. Discussions with teammates, faculty members, and project guides helped improve my understanding of analytical concepts and system development. I learned how to explain educational data patterns, machine learning results, and dashboard

insights in a clear and understandable manner.

5.3 Application of Technical Principles

The project applied relevant Data Science and analytical principles such as data preprocessing, data integration, classification, clustering, visualization, and responsible data management. The use of Data Warehouse, Text Mining, Decision Tree, K-Means, KNIME, WEKA, and Power BI provided a structured approach for analyzing educational information.

These practices helped improve the reliability, usability, and practical relevance of the proposed Educational Learning Analytics system.

5.4 Insights into the Industry

The project provided valuable insights into how educational institutions and organizations can use data analytics for improved decision-making. I learned that educational data can provide meaningful information about student performance, learning behavior, and academic trends.

The project also demonstrated the importance of combining Machine Learning, Data Mining, and Business Intelligence tools. This experience helped me understand practical applications related to Data Science, Artificial Intelligence, Educational Technology, and Business Analytics.

5.5 Conclusion of Personal Development

This project was a valuable learning experience that enhanced my academic knowledge, technical skills, problem-solving abilities, and professional development. It strengthened my understanding of Data Warehousing, Text Mining, Machine Learning, Clustering, KNIME, WEKA, and Microsoft Power BI.

Through this work, I developed greater confidence in data analysis, machine learning, visualization, research, and technical communication. Overall, the project helped connect academic knowledge with practical applications of Educational Learning Analytics.

CHAPTER 6 – CONCLUSION

6.1 Summary of Key Findings

The project “Educational Learning Analytics using Data Warehouse, Text Mining, Decision Tree, K-Means Clustering, KNIME, WEKA and Microsoft Power BI” successfully demonstrated the application of Data Analytics and Machine Learning techniques for analyzing educational information.

The system uses a Data Warehouse to organize educational data and applies Text Mining to analyze textual information. Decision Tree techniques support student performance classification, while K-Means Clustering helps group students with similar characteristics. KNIME and WEKA support analytical processing and experimentation, while Microsoft Power BI provides visual dashboards for presenting the results.

The project demonstrates that Educational Learning Analytics can reduce manual analysis and provide educators with a clearer understanding of student performance and learning behavior.

6.2 Impacts and Significance

The project highlights the importance of data-driven analysis in modern education. By organizing and analyzing student information, the system can help educators and administrators understand academic performance more efficiently.

The proposed system can support:

- Student performance analysis.
- Educational data management.
- Student classification.
- Learning pattern identification.
- Student grouping and segmentation.
- Feedback analysis.
- Improved educational decision-making.
- Interactive academic reporting.

The project demonstrates how Data Warehousing, Text Mining, Machine Learning, KNIME, WEKA, and Microsoft Power BI can be combined to solve practical educational data analysis problems.

6.3 Future Prospects

The project can be extended with several advanced features in the future. These may include real-time educational data analysis, automatic alerts for low-performing students, personalized learning recommendations, predictive academic performance analysis, and integration with online learning platforms. Advanced Artificial Intelligence techniques could also be integrated to improve student performance prediction, analyze learning behavior, and provide intelligent recommendations. Integration with additional educational systems and cloud-based analytics platforms could further improve the usefulness of the proposed system.

REFERENCES

- Lou, Y. (2025). Performance prediction using educational data mining: A systematic review of machine learning approaches. *Discover Computing*.
- Malik, S., et al. (2025). Advancing educational data mining for enhanced student performance prediction. *Scientific Reports*.
- Jin, J., et al. (2025). Student behavior patterns in vocational education big data using K-means clustering. *Discover Artificial Intelligence*.
- Alzahrani, N., et al. (2025). Identifying weekly student engagement patterns in E-learning using clustering techniques. *Electronics*, 14(15), 3018.
- Islam, M. M., et al. (2025). The integration of explainable AI in Educational Data Mining. *International Journal of Educational Research Open*.
- Jiang, R., et al. (2025). Educational applications based on diagnostic learning analytics and big data mining. *Computers and Education: Artificial Intelligence*.
- Križanić, S. (2025). Data and process mining in analysing student behaviour. *INDECS*.
- Automated Classification and Sentiment Analysis of Student Feedback Using Text Mining. (2025). *International Journal of Information and Education Technology*.
- Bataineh, A., et al. (2025). An Empirical Study on Adaptive Learning Using Decision Tree Models. *IEEE*.
- Clustering and Personalized Course Recommendation Using K-Means for Educational Data. (2025). *ACM International Conference Proceedings*

APPENDICES

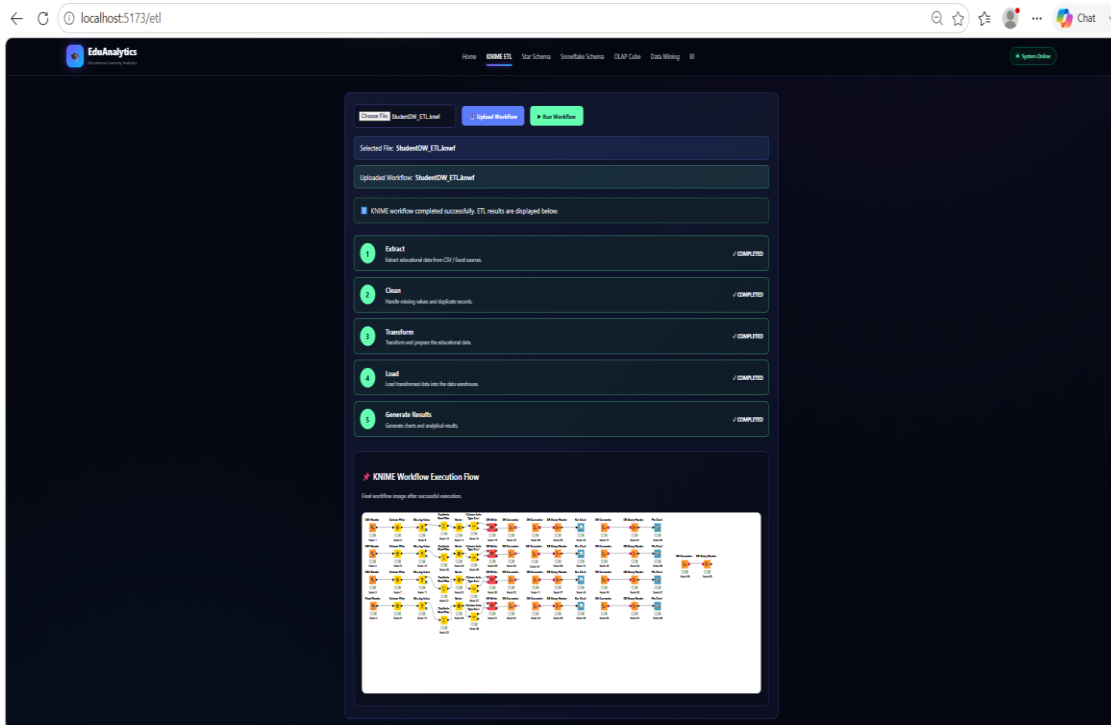


FIG 01 : KNIME WORKFLOW

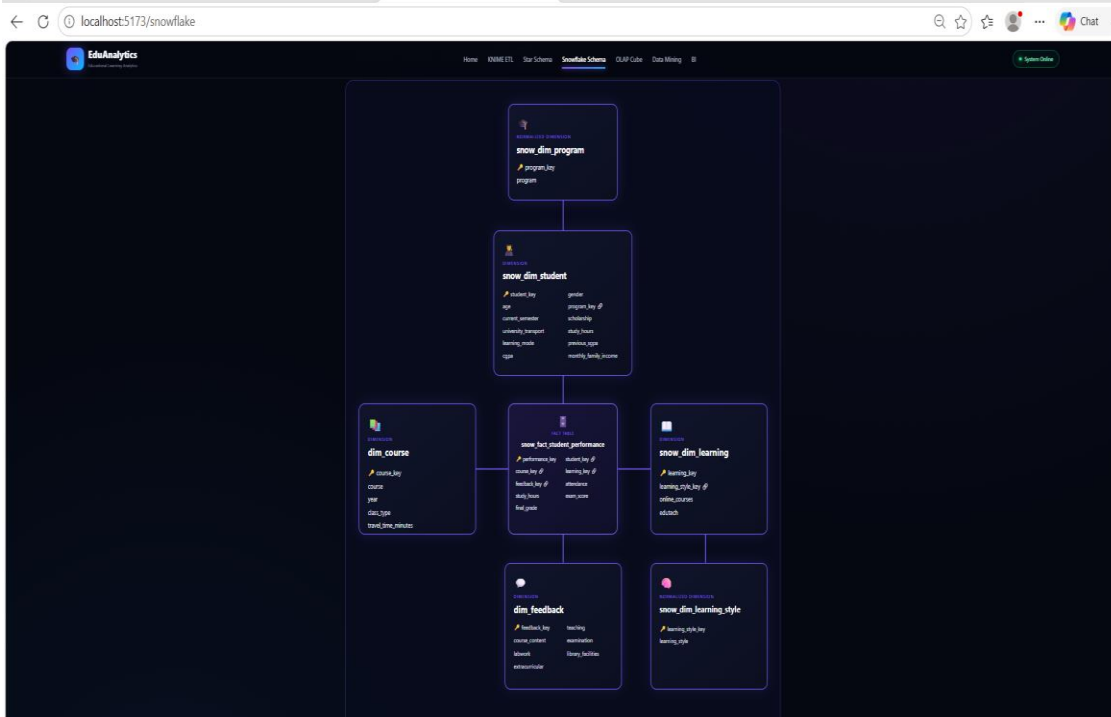


FIG 02 : SNOWFLAKE SCHEMA

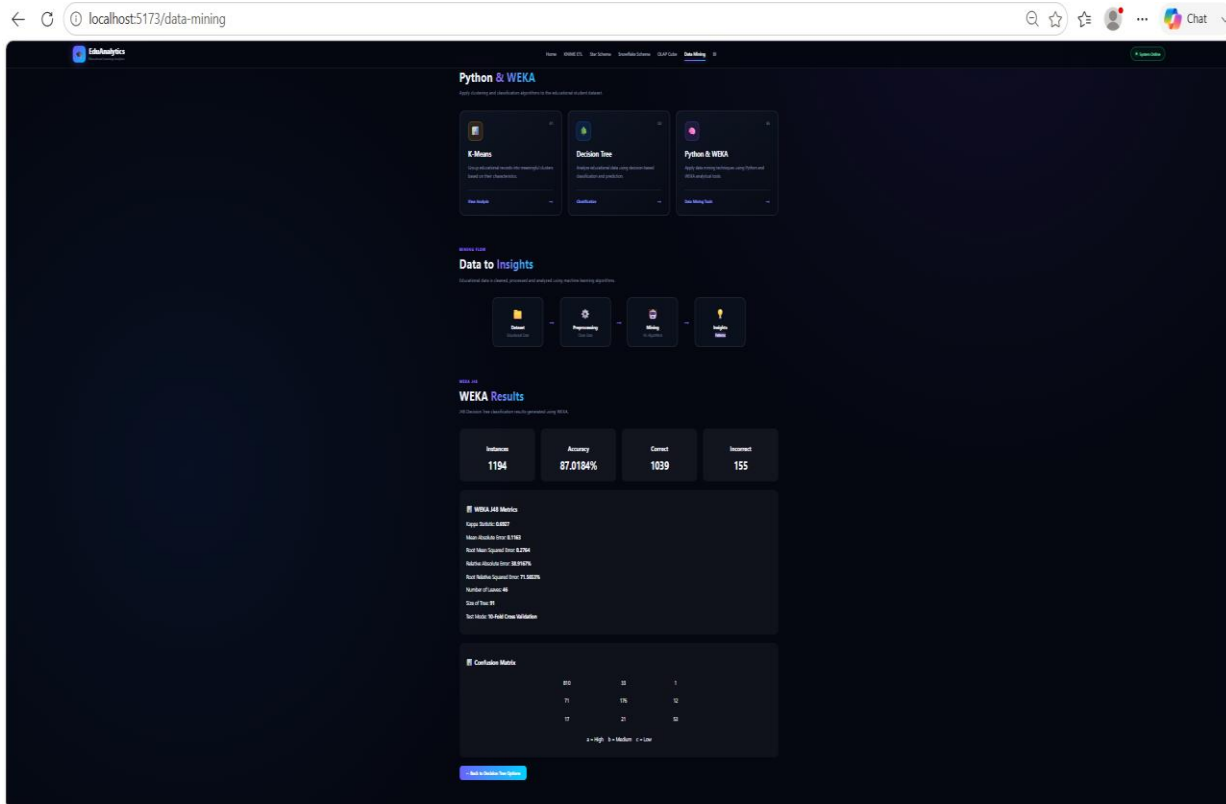


FIG 3 : WEKA DECISION TREE (J48)

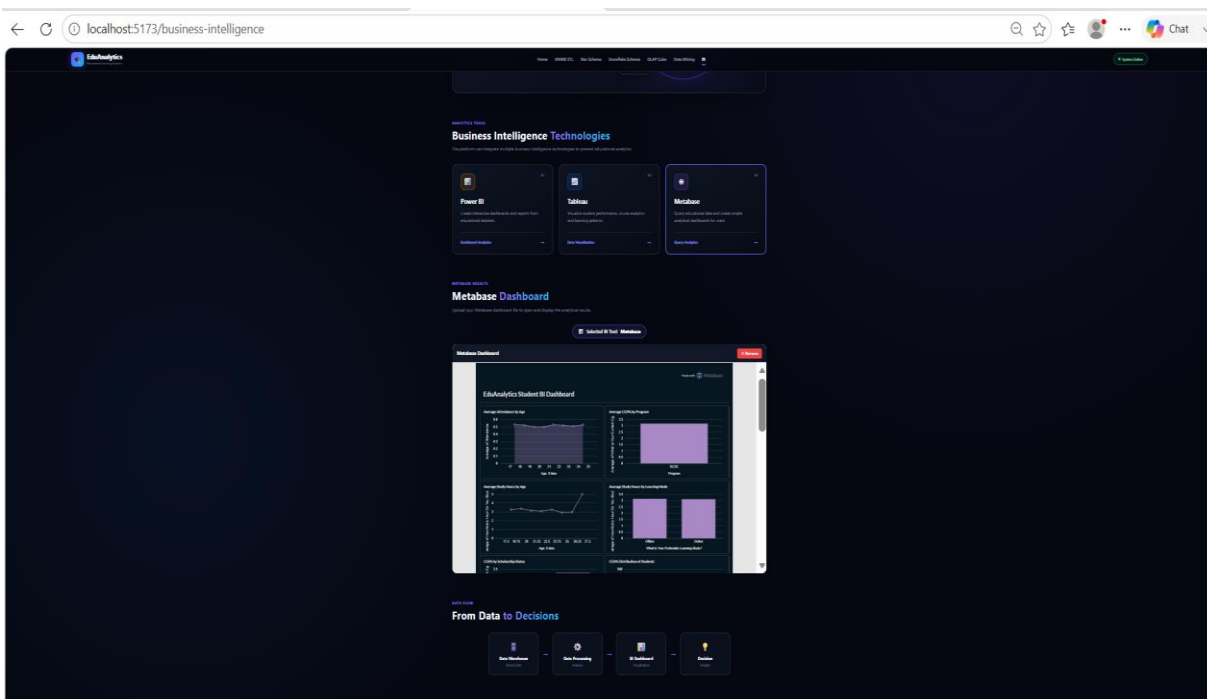


FIG 4: METABASE DASHBOARD